



Predicting Customer Churn From SyriaTel

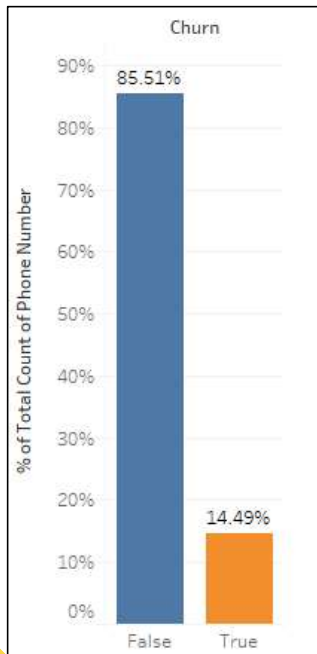
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Problem Statement

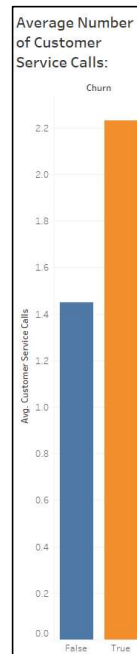


- Customer Retention remains a critical KPI For SyriaTel
- Acquiring new customers is often more expensive than keeping existing ones.
- SyriaTel is experiencing customer churn where users discontinue their service.
- This project seeks to build a classifier to predict churn
- Identifying the patterns and factors that drive churn can help the company take proactive measures to reduce customer loss, enhance loyalty, and increase revenue.

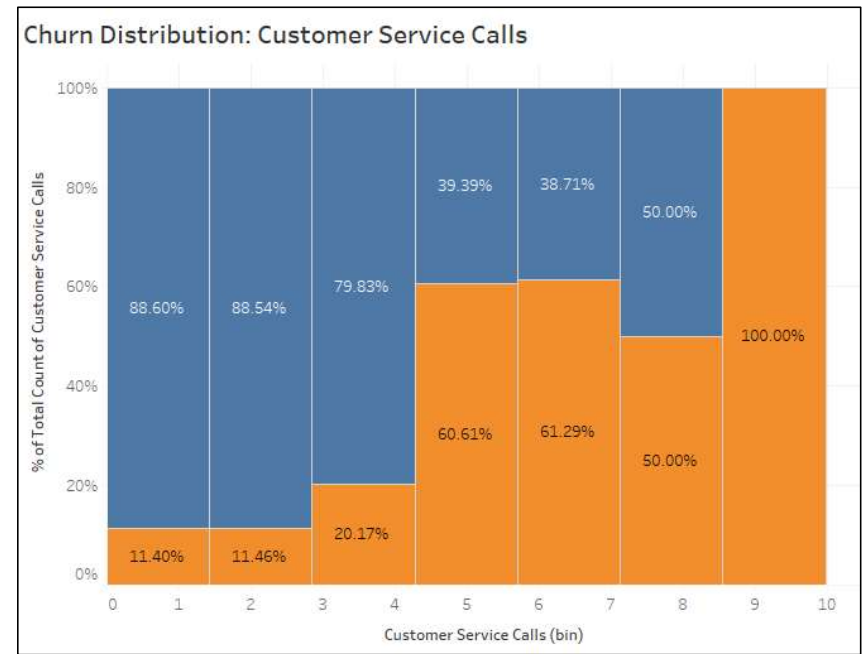
EDA Sample - Customer Service Calls



14.49% Churn Rate at SyriaTel



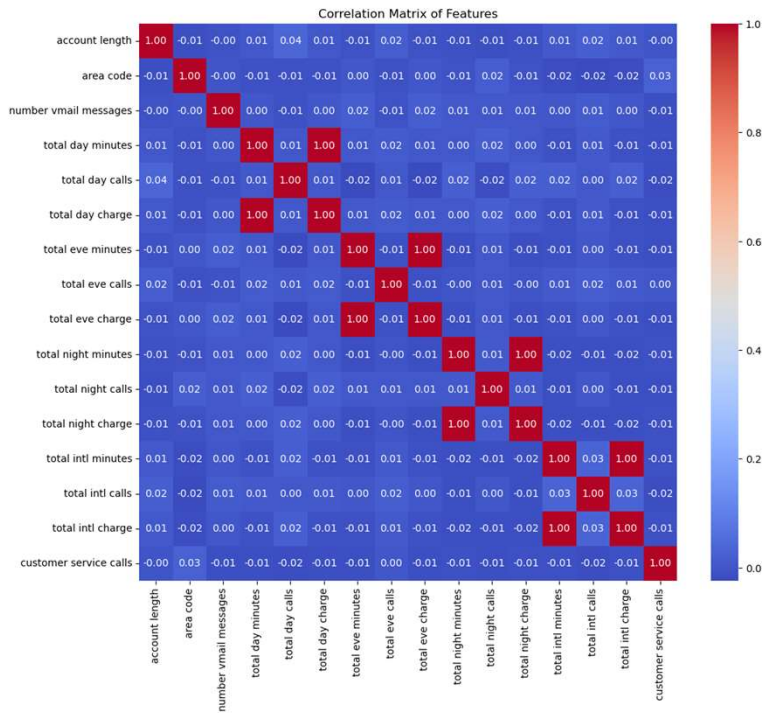
On average, churned customers made more than 2 customer service calls



For customers who made more than 4 Customer service calls, the churn probability is very high

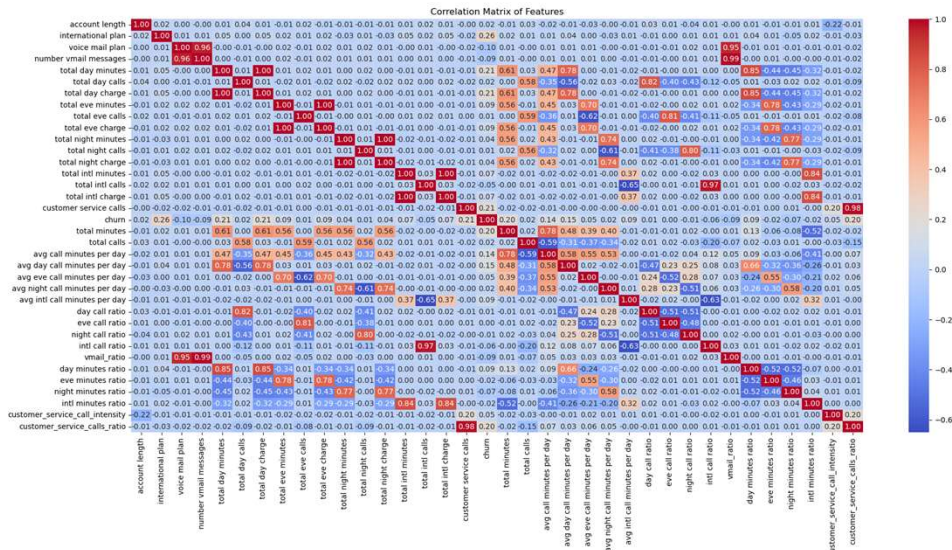
There could be underlying problems not being solved leading to high churn rate

EDA Sample – Correlation



- The relationship between the original features appears non-linear
 - Except for charges vs call-time
- Logistic Regression does not handle non-linear relationships well

EDA Sample - Correlation



Feature Engineering:

Feature Engineering was used to try and make better sense of the data. The following features were derived:

- Total Minutes = total day minutes + total eve minutes + total night minutes
- Total Calls = total day calls + total eve calls + total night calls
- Avg Call Minutes Per Day = total minutes / total calls
- Avg Day Call Minutes Per Day = total day minutes / total day calls
- Avg Eve Call Minutes Per Day = total eve minutes / total eve calls
- Avg Night Call Minutes Per Day = total night minutes / total night calls
- Avg Intl Call Minutes Per Day = total intl minutes / total intl calls
- Day Call Ratio = total day calls / total calls
- Eve Call Ratio = total eve calls / total calls
- Night Call Ratio = total night calls / total calls
- Intl Call Ratio = total intl calls / total calls
- Vmail Ratio = number vmail messages / total calls
- Day Minutes Ratio = total day minutes / total minutes
- Eve Minutes Ratio = total eve minutes / total minutes
- Night Minutes Ratio = total night minutes / total minutes
- Intl Minutes Ratio = total intl minutes / total minutes
- Customer Service Call Intensity = customer service calls / account length
- Customer Service Calls Ratio = customer service calls / total calls

- Feature Engineering attempted to explore potentially higher correlations

Modeling – Finding the Best Model



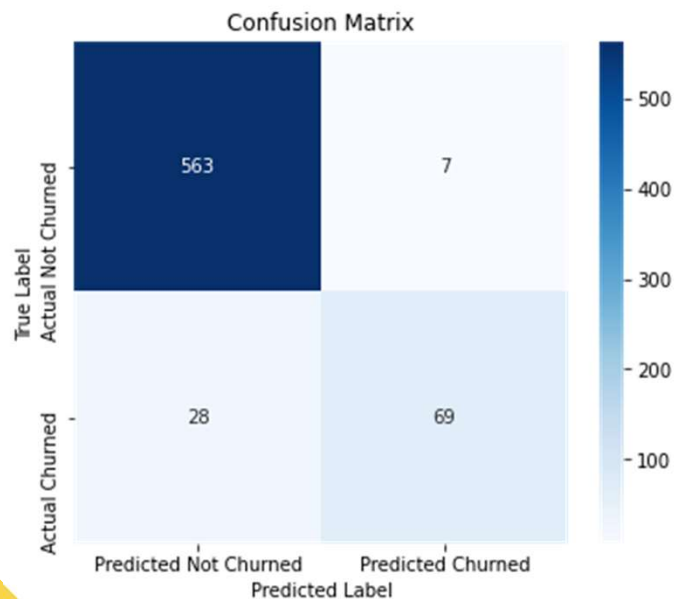
Model Iteration	Accuracy	Precision	Recall	F1 Score	AUC-ROC
Baseline - Random Forest	94.15%	90.28%	67.01%	76.92%	82.89%
Engineered - Random Forest	93.25%	93.33%	57.73%	71.34%	78.52%
Engineered Only - Random Forest	92.05%	92.31%	49.48%	64.43%	74.39%
Baseline - Support Vector Machine	90.85%	84.62%	45.36%	59.06%	71.98%
Engineered Only - Decision Tree	90.70%	67.68%	69.07%	68.37%	81.73%
Engineered - Support Vector Machine	90.70%	81.82%	46.39%	59.21%	72.32%
Baseline - Decision Tree	90.55%	69.32%	62.89%	65.95%	79.07%
Engineered Only - Support Vector Machine	90.55%	80.36%	46.39%	58.82%	72.23%
Engineered Only - K Nearest Neighbour	89.66%	75.93%	42.27%	54.30%	69.99%
Engineered - Decision Tree	89.36%	63.27%	63.92%	63.59%	78.80%
Engineered Only - Stochastic Gradient Descent	89.21%	71.19%	43.30%	53.85%	70.16%
Baseline - K Nearest Neighbour	88.61%	69.09%	39.18%	50.00%	68.10%
Engineered - AdaBoost Regression	88.16%	65.52%	39.18%	49.03%	67.83%
Engineered - K Nearest Neighbour	88.16%	66.67%	37.11%	47.68%	66.98%
Engineered Only - AdaBoost Regression	87.26%	63.04%	29.90%	40.56%	63.46%
Engineered - Ridge Regression	87.26%	73.08%	19.59%	30.89%	59.18%
Baseline - AdaBoost Regression	86.81%	57.38%	36.08%	44.30%	65.76%
Engineered - Logistic Regression	86.36%	55.77%	29.90%	38.93%	62.93%
Engineered Only - Logistic Regression	86.21%	55.56%	25.77%	35.21%	61.13%
Baseline - Ridge Regression	86.06%	60.00%	12.37%	20.51%	55.48%
Engineered Only - Ridge Regression	86.06%	60.00%	12.37%	20.51%	55.48%
Baseline - Logistic Regression	85.91%	53.33%	24.74%	33.80%	60.53%
Baseline - Stochastic Gradient Descent	85.01%	38.46%	5.15%	9.09%	51.88%
Engineered - Stochastic Gradient Descent	84.11%	33.33%	9.28%	14.52%	53.06%

To also explore the effect of different feature combinations in the model performance, the following 3 dataframes were used:

- **Baseline:** Here, the data used to train and evaluate the models only have the **original features**
- **Engineered:** Here, the data used to train and evaluate the models have both the **original features** and **all the engineered features**
- **Engineered Only:** Here, the data used to train and evaluate the models have **all the engineered features**, but some of the original features have been removed

- Random Forest Model with the baseline data yielded the highest Accuracy, Recall, F1 and AUC-ROC scores

Modeling – Feature Selection



- Refining feature selection improved the model performance slightly
- This was done through a function that randomly selected 10 features 10000 times and evaluated the best performing one

X = ['total night charge', 'total intl charge', 'voice mail plan', 'total eve minutes', 'total eve charge', 'total day charge', 'total night calls', 'total intl calls', 'customer service calls', 'international plan']

The performance metrics improved to:

- Accuracy: 0.950525
- Recall: 0.711340
- Precision: 0.932432
- F1-score: 0.807018

Conclusions and Recommendations



- The recall is still a bit low. Other classification techniques e.g. XGBoost Classification could be attempted to yield better model performance.
- Other ML techniques like SMOTE can be used to handle the data imbalance and subsequently enhance recall
- More descriptive features could improve model performance e.g. Customer Service Calls Reasons – Addressing the core challenges that necessitate Customer Complaints will reduce churn
- Improving the customer experience of customers with international call plan will reduce the rate of churn

Thank you

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